GeoSoil-Cuba

Integrating Soil Mapping, Land Cover, and Sentinel-2 Imagery for Geospatial Analysis in Cuba

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October 2026

Abstract

GeoSoil-Cuba is an open-source geospatial analysis project that integrates soil mapping, land-cover information, and Sentinel-2 satellite imagery within a reproducible Python-based workflow. The study focuses on Jaruco, Mayabeque Province, Cuba, and combines soil-map preparation and spatial validation with satellite-derived vegetation monitoring. A 1:25,000-scale soil dataset was evaluated within an Area of Interest (AOI) covering approximately 178.47 km^2. Soil polygons covered 162.85 km^2 (91.25% of the AOI), while 4.20 km^2 (2.35%) remained uncovered outside the mapped water-related gaps. Topological assessment identified 90 pairs of soil polygons with positive-area overlaps, highlighting issues requiring further validation. For vegetation monitoring, Sentinel-2 Level-2A imagery acquired between November 2025 and June 2026 was processed to calculate the Normalized Difference Vegetation Index (NDVI). Fifteen acquisitions met a regional cloud-coverage criterion of less than 10% within a unified agricultural region of approximately 16.22 km^2. Mean NDVI declined from 0.688 in November 2025 to 0.466 in March 2026, before increasing to 0.737 in June 2026. A complementary analysis examined NDVI variability in three locally verified potato fields using circular sampling polygons digitized in QGIS. The results demonstrate the utility of integrating historical soil information, land-cover data, and satellite observations in a documented geospatial workflow. However, the observed vegetation-index patterns do not establish causal relationships between soil properties, crop development, and agricultural management.

Keywords: Geospatial Analysis; Soil Mapping; Remote Sensing; Sentinel-2; NDVI; Python; GIS.

1 Introduction

Soil characteristics and vegetation conditions are closely related components of terrestrial ecosystems. Understanding their spatial relationships is important for agricultural assessment, land-resource management, and environmental monitoring. Geographic Information Systems (GIS) and satellite remote sensing provide complementary sources of information for investigating these relationships across different spatial and temporal scales.

Conventional soil maps describe the spatial distribution of soil types and associated characteristics. However, integrating these datasets with contemporary geospatial information may require coordinate reference system harmonization, geometry validation, spatial clipping, attribute interpretation, and assessment of mapping completeness. These operations are particularly

relevant when working with historical cartographic datasets that were originally developed using formats and classification systems different from those commonly employed in modern geospatial workflows.

Satellite observations complement soil mapping by providing repeated measurements of land-surface conditions. In particular, the Sentinel-2 mission supplies multispectral imagery suitable for vegetation monitoring. The Normalized Difference Vegetation Index (NDVI), calculated from red and near-infrared reflectance, provides an indirect indicator of vegetation greenness and canopy development. Temporal NDVI observations can characterize changes in vegetation conditions, although their interpretation depends on land cover, acquisition quality, and agricultural context.

GeoSoil-Cuba is an open-source geospatial analysis project developed to integrate soil mapping, land-cover information, and Sentinel-2 satellite imagery within a reproducible Python-based workflow. The project focuses on a study area in Jaruco, Mayabeque Province, Cuba, using a 1:25,000-scale soil map as the primary spatial framework for subsequent soil–vegetation analysis.

The workflow has two complementary analytical components. The first prepares and evaluates the soil dataset through geometry validation and repair, coordinate reference system harmonization, spatial intersection and clipping, classification interpretation, and soil-map coverage assessment. Water-body information helps distinguish areas excluded from terrestrial soil mapping from unresolved gaps in the source dataset. The second component processes Sentinel-2 observations to examine temporal NDVI variability in a selected agricultural region and in three locally verified potato-field samples. Together, these components provide spatially organized datasets and exploratory vegetation observations without presuming a causal relationship between mapped soil classes and vegetation dynamics.

The main objectives of this study are to:

1. Prepare and validate the existing soil cartographic information for geospatial analysis within the selected Area of Interest (AOI).

2. Evaluate the spatial completeness and classification consistency of the soil dataset, distinguishing mapped water bodies from unresolved soil-data gaps.

3. Integrate the processed soil information with land-cover classifications and organize it for analysis alongside Sentinel-2 observations.

4. Process Sentinel-2 imagery and characterize temporal NDVI variability within a selected agricultural region.

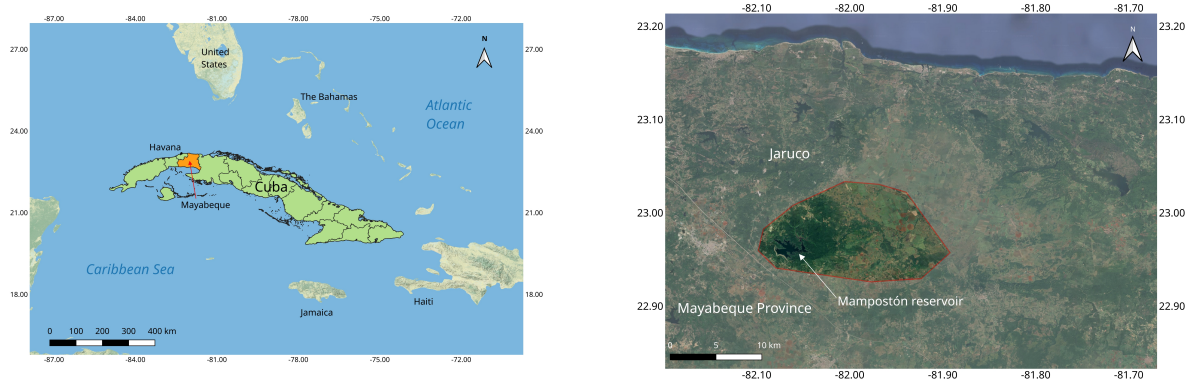
5. Compare NDVI observations from three locally verified potato-field samples, recognizing the limits of inferring crop development from satellite reflectance alone.

The contribution of GeoSoil-Cuba lies primarily in the integration, validation, and documentation of heterogeneous geospatial datasets through an open-source analytical workflow. Rather than proposing a new soil classification system or establishing causal relationships between soil properties and vegetation performance, the project provides a documented framework for exploratory soil–vegetation analysis that can be extended through additional satellite observations, field measurements, and spatial datasets.

2 Study Area

The GeoSoil-Cuba study area is located in Jaruco, Mayabeque Province, western Cuba. The Area of Interest (AOI) covers approximately 178.47 km^2 and encompasses agricultural landscapes, natural and semi-natural vegetation, and the Mampostón reservoir (Figure 1). Administrative boundaries were obtained from the GADM database, version 4.1, using the Cuba GeoPackage dataset (`gadm41_CUB.gpkg`) (GADM [2022]). Both location maps were prepared in QGIS. Cartographic elements, including coordinate grids, scale bars, north arrows, and geographic labels, were configured using the QGIS print layout.

This geographic setting provides a basis for integrating historical soil cartography, land-cover information, and satellite-derived vegetation observations.



(a) Geographic location of Mayabeque Province, Cuba.

(b) Study area in Jaruco, Mayabeque Province.

Figure 1: Geographic location of the study area: (a) Mayabeque Province within Cuba and (b) study area in Jaruco Municipality, Mayabeque Province. Maps prepared by the author using QGIS. Administrative boundaries derived from GADM version 4.1 (GADM [2022]).

The AOI was defined using a geographic boundary provided in Keyhole Markup Language (KML) format. The original boundary uses the WGS 84 geographic coordinate reference system (EPSG:4326). Its approximate geographic extent ranges from 22.925 85° to 23.033 32° north latitude and from −82.098 37° to −81.892 15° longitude.

For spatial analyses involving the historical soil map, the AOI was transformed to NAD27 / Cuba Norte (EPSG:3795), the projected coordinate reference system used by the source cartographic dataset. This transformation facilitates spatial intersection, clipping, and area calculations within a common coordinate reference system.

The Mampostón reservoir is a relevant geographic feature within the AOI. Because open-water surfaces are not expected to be represented by terrestrial soil mapping units, their presence must be considered when assessing soil-map coverage. Accordingly, areas without mapped soil information must be distinguished from water surfaces and from other gaps that may remain within the terrestrial analysis area.

Within the broader AOI, a unified agricultural region was delineated for the regional NDVI time-series analysis. A complementary analysis examined vegetation-index observations within three locally verified potato fields, represented by sampling polygons digitized in QGIS. These spatial units support different analytical purposes: the AOI provides the overall context for soil and land-cover integration, whereas the agricultural region and field samples support the examination of vegetation variability at regional and field scales, respectively.

The study area was selected to evaluate the integration and spatial consistency of existing soil cartographic information, land-cover classifications, and satellite observations. The combined analyses provide a basis for assessing soil-map coverage and examining temporal vegetation-index patterns without assuming that observed NDVI differences are caused by soil properties or agricultural management.

3 Data Sources and Technical Environment

This study integrates existing soil and ancillary spatial datasets with Sentinel-2 satellite imagery to examine soil-map coverage, vegetation distribution, and temporal vegetation dynamics within the study area which were obtained from the Sentinel-2 Level-2A product collection through the Copernicus Data Space Ecosystem (CDSE) [Drusch et al. (2012)]. The workflow combines vector-data preparation and spatial analysis with satellite image processing and NDVI-based assessment.

The principal datasets, their spatial characteristics, and their roles in the analysis are summarized in Table 1.

Table 1: Summary of spatial and satellite data sources used in the GeoSoil-Cuba workflow.

Dataset	Source or format	Spatial reference	Role in the study
Soil map	Faculty of Geography, University of Havana; MapInfo TAB	NAD 27 Cuba datum; Lambert Conformal Conic; metres	Delineation and analysis of mapped soil units within the study area. Original scale: 1:25,000.
Area of Interest (AOI)	KML boundary	WGS 84 geographic coordinates (EPSG:4326)	Defines the study-area boundary and spatial extent of the analysis.
Water bodies	Faculty of Geography, University of Havana; MapInfo TAB	As supplied; harmonized for spatial analysis	Identification of water surfaces and interpretation of apparent gaps in soil-map coverage.
Vegetation and land cover	Faculty of Geography, University of Havana; vector dataset	As supplied; harmonized for spatial analysis	Integration of soil units with vegetation and land-cover attributes.
Sentinel-2 imagery	Copernicus Data Space Ecosystem; Level-2A products	Tiles T17QLF and T17QMF; bands B04 and B08 at 10 m resolution; SCL at 20 m	Satellite-based vegetation analysis, including NDVI calculation and temporal assessment. Search period: November 2025 to June 2026.

3.1 Soil and Ancillary Spatial Data

The primary soil dataset consists of a 1:25,000-scale soil map of Mayabeque Province, Cuba, distributed in MapInfo format. The dataset contains polygon geometries representing mapped soil units and associated attributes describing soil types and subtypes according to the classification

system used in the source database.

The soil map uses the NAD27 / Cuba Norte projected coordinate reference system (EPSG:3795). Its original projection definition was examined and standardized before spatial processing.

The Area of Interest (AOI) was defined using a polygon provided in Keyhole Markup Language (KML) format, originally referenced to WGS 84 (EPSG:4326). The AOI was transformed to the soil map's projected coordinate reference system for the relevant spatial operations.

An additional water-body dataset, also supplied in MapInfo format, was incorporated to identify the Mampostón reservoir and distinguish mapped water surfaces from areas lacking soil-map coverage.

The original spatial datasets were preserved unchanged. Derived geometries and attributes were stored separately to maintain traceability throughout the analytical workflow.

The original production dates and authorship of the soil and ancillary datasets have not been independently verified. Their spatial and thematic interpretation must therefore account for the characteristics and limitations of the supplied source data.

3.2 Vegetation and Land-Cover Data

Vegetation and land-cover information was obtained from an existing vector dataset containing categorical descriptions of vegetation and land-cover units.

The dataset includes the attributes `LCNAME`, `NEWVEG`, and `TNC_CLASS`, which provide complementary classification information.

These attributes were examined and incorporated into the spatial integration workflow to characterize vegetation and land-cover categories associated with the mapped soil units.

The vegetation dataset was processed independently from the Sentinel-2 imagery. Its categorical classifications therefore represent information from an existing spatial dataset rather than land-cover classes derived from the satellite observations analyzed in this study.

The production date and original classification methodology of the vegetation dataset have not been independently verified. Consequently, its temporal correspondence with the 2025–2026 satellite observations cannot be assumed.

The soil, water-body, and vegetation-cover datasets were provided courtesy of the Faculty of Geography, University of Havana, Cuba (Facultad de Geografía, Universidad de La Habana), with permission for research, analysis, and public redistribution. The institution is acknowledged as the data provider; the original production dates and authorship of these datasets have not been independently verified.

3.3 Sentinel-2 Satellite Imagery

Multispectral satellite observations were obtained from the Sentinel-2 Level-2A product collection through the Copernicus Data Space Ecosystem (CDSE).

Satellite products were discovered using the SpatioTemporal Asset Catalog (STAC) interface, which supports searches based on geographic extent, acquisition date, and product metadata.

The catalogue search covered the period from November 2025 through June 2026. Products associated with Sentinel-2 tiles T17QLF and T17QMF were considered for the study area.

The analysis focused on the red (B04) and near-infrared (B08) spectral bands, both provided at a spatial resolution of 10 m. The Scene Classification Layer (SCL), available at a spatial

resolution of 20 m, was used to support quality screening and the assessment of potential cloud and shadow contamination.

The initial satellite inventory contained 226 observations. Following image processing, 32 NDVI rasters were prepared for the selected agricultural analysis region. A subsequent regional cloud-screening procedure retained 15 acquisitions satisfying the criterion of less than 10% cloud coverage within that region.

This regional criterion reduced the influence of cloud contamination but did not establish that all pixels in the retained observations were free of clouds, haze, or shadows.

Detailed acquisition selection, radiometric correction, quality screening, and NDVI calculation procedures are described in Section 4.

3.4 Software and Technical Environment

GeoSoil-Cuba was developed using an open-source Python geospatial environment managed through Conda, with packages obtained primarily from the `conda-forge` channel.

The documented environment uses Python 3.12.14 and GDAL 3.13.3.

Vector-data processing was performed using GeoPandas 1.1.4 and Shapely 2.1.2, while coordinate reference system transformations were supported by PyProj 3.8.0.

Raster processing was supported by Rasterio 1.5.1, including the handling of georeferenced satellite imagery and derived NDVI products.

NumPy 2.5.3 and Pandas 3.0.5 provided numerical and tabular data processing capabilities, respectively. Matplotlib 3.11.1 was used to generate analytical figures and visualizations.

Sentinel-2 catalogue discovery was supported by PySTAC Client 0.9.0, while Requests 2.34.2 provided HTTP communication functionality.

The analytical workflow was organized into Jupyter notebooks, with JupyterLab 4.6.3 included in the documented environment. QGIS 3.44.14-Solothurn was also used for the visual selection and digitization of the three potato-field sampling polygons.

The Conda environment specification and project source code provide a record of the principal software dependencies used in the analysis. The software versions reported here correspond to the documented environment and should be verified against the final environment specification before publication.

3.5 Data Characteristics and Limitations

The integration of historical soil cartography, existing vegetation classifications, and satellite observations introduces differences in spatial representation, classification structure, and temporal coverage.

The soil dataset represents mapped soil units rather than direct measurements of contemporary soil conditions. Its interpretation is therefore constrained by the characteristics of the source cartographic dataset, including its original scale, soil classification system, and spatial completeness. The dataset uses a Lambert Conformal Conic projection referenced to the NAD 27 Cuba datum, with coordinates expressed in metres. The original map scale and classification metadata should be considered when interpreting the spatial detail and thematic meaning of the mapped soil units.

The vegetation dataset provides categorical land-cover information, whereas Sentinel-2 observations describe spectral conditions at specific acquisition dates. Because the original production

date of the vegetation dataset has not been verified, these sources cannot be assumed to represent the same period.

Satellite-derived vegetation indices may also be affected by atmospheric conditions, residual cloud or shadow contamination, land-cover heterogeneity, radiometric processing choices, and the spatial resolution of the imagery.

In addition, radiometric parameters were not independently verified for every processed Sentinel-2 product. The use of common parameters for products with unavailable metadata is therefore a methodological assumption that should be considered when interpreting the derived NDVI values.

These characteristics were considered when preparing the datasets and interpreting the spatial coverage results and temporal NDVI patterns. The combined workflow supports geospatial integration and exploratory vegetation monitoring but does not, by itself, establish causal relationships between soil properties, vegetation conditions, and agricultural management.

4 Methodology

The GeoSoil-Cuba workflow integrates soil-map preparation, vegetation and land-cover information, and Sentinel-2 satellite observations through a sequence of spatial processing and analytical stages. The methodology addresses soil-map coverage and topology, integration of soil and vegetation attributes, satellite-image processing, and temporal analysis of the Normalized Difference Vegetation Index (NDVI).

4.1 Workflow Overview

The main processing stages and their sequence are illustrated in fig. 2.

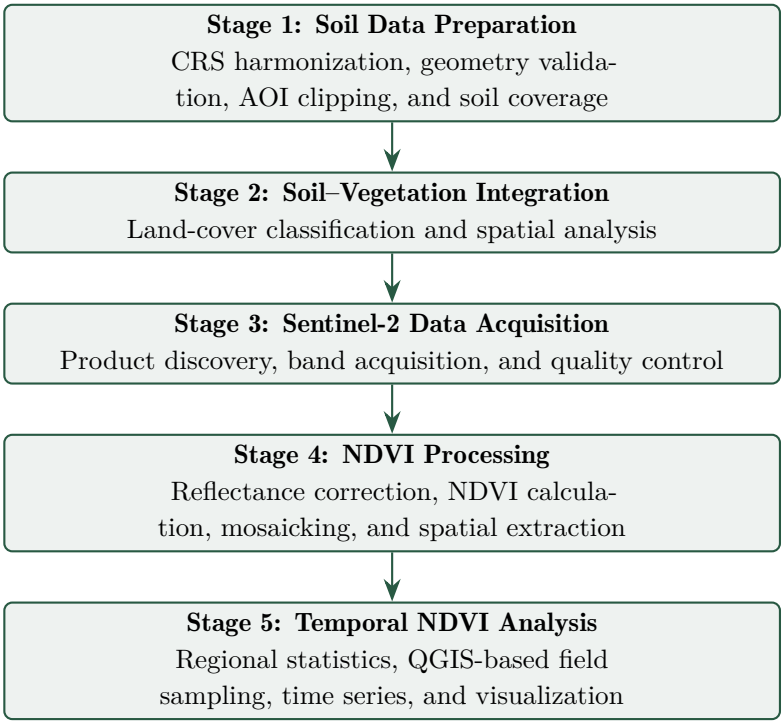


Figure 2: Simplified GeoSoil-Cuba workflow showing the principal stages of spatial data preparation, integration, and NDVI analysis.

The workflow comprises five principal stages: soil-data inspection and preparation; soil-map intersection with the Area of Interest (AOI); integration of soil and vegetation information; acquisition and quality screening of Sentinel-2 imagery; and calculation and analysis of NDVI.

The stages were implemented through a series of Python notebooks, with QGIS used for the visual selection of agricultural sampling areas. Intermediate spatial datasets, tables, and figures were exported to support quality control, reproducibility, and subsequent analysis.

4.2 Soil Data Preparation and AOI Intersection

The Mayabeque soil map at a scale of 1:25,000 was inspected before spatial analysis. The soil dataset and the AOI were evaluated for their coordinate reference systems and geometry characteristics.

The AOI, originally supplied in WGS 84 geographic coordinates (EPSG:4326), was transformed to NAD27 / Cuba Norte (EPSG:3795), the projected coordinate reference system used for soil-map analysis.

Soil geometries were checked for validity. Invalid geometries were repaired using `make_valid()`, and resulting geometry collections were normalized to polygonal geometries where necessary. The original source data were preserved, and repaired geometries were stored separately for analysis.

Soil features intersecting the AOI were selected and clipped to its boundary. The resulting dataset was used to assess soil-map coverage and calculate the areas represented by mapped soil units.

Spatial differences between the AOI and the mapped soil coverage were examined alongside the water-body dataset to distinguish reservoir-related areas from other uncovered locations.

Areas not represented by soil polygons and not explained by the mapped reservoir were retained as unresolved data gaps. No soil classifications were assigned to these areas without supporting information.

The resulting soil dataset was also examined for undefined classification attributes and overlapping polygon geometries. These conditions were documented as data-quality findings. Positive-area overlaps were identified but were not automatically eliminated by modifying the original soil-unit boundaries.

4.3 Soil–Vegetation Integration

The processed soil layer was integrated with existing vegetation and land-cover information to support spatial analysis of mapped soil units in their vegetation context.

The attributes `LCNAME`, `NEWVEG`, and `TNC_CLASS` were retained for analysis. Missing or undefined classification values were identified during data preparation rather than being replaced with unsupported categories.

The integrated vector dataset provides a common spatial framework for examining the distribution of vegetation classifications across mapped soil units. It also supports the preparation of thematic maps and the examination of spatial coverage and classification completeness.

The resulting dataset was exported in GeoPackage format for subsequent analysis and cartographic representation. The integration of these datasets does not, by itself, establish statistical or causal relationships between soil properties and vegetation conditions.

4.4 Sentinel-2 Acquisition and Quality Screening

Sentinel-2 Level-2A imagery was obtained through the Copernicus Data Space Ecosystem (CDSE) for the period November 2025 to June 2026. Products were discovered through the SpatioTemporal Asset Catalog (STAC) interface using the geographic extent and acquisition period relevant to the study area.

The study area intersects Sentinel-2 tiles T17QLF and T17QMF. The red band (B04) and near-infrared band (B08), both provided at 10 m spatial resolution, were selected for NDVI calculation.

Acquisition records were organized by date, satellite, and tile to identify complete B04/B08 pairs. Incomplete band pairs were excluded from NDVI processing.

The Scene Classification Layer (SCL), provided at 20 m spatial resolution, was used to support quality screening for potentially cloud- and shadow-contaminated observations.

The initial satellite inventory contained 226 observations. Following tile-level processing and preparation of the agricultural analysis region, 32 acquisition dates were represented by analysis-area NDVI rasters. A subsequent regional cloud-screening procedure retained 15 acquisitions meeting the final criterion of less than 10% cloud coverage within the agricultural analysis region.

The cloud-coverage threshold was used as an acquisition-selection criterion. It does not guarantee that all pixels in the retained observations are free from clouds, haze, or shadows. Cloud screening and numerical NDVI validation were therefore treated as distinct quality-control procedures.

Regional cloud coverage was calculated using SCL classes 8, 9, and 10, representing medium-probability clouds, high-probability clouds, and thin cirrus, respectively.

Cloud shadows, represented by SCL class 3, were evaluated separately and were not included in the reported cloud-coverage percentage.

4.5 Radiometric Correction and NDVI Calculation

Sentinel-2 Level-2A spectral bands are distributed as integer digital numbers (DN). Bottom-of-atmosphere (BOA) reflectance was calculated using the radiometric quantification value and additive offset associated with the satellite products:

$$\rho_{\mathrm{BOA}} = \frac{DN + O_{\mathrm{BOA}}}{Q_{\mathrm{BOA}}}, \tag{1}$$

where DN represents the stored pixel value, O_{BOA} is the additive radiometric offset, and Q_{BOA} is the reflectance quantification value.

An examination of 111 available Sentinel-2 product XML metadata files identified a quantification value of 10,000 and additive offsets of -1000 for both B04 and B08. The examined products corresponded to processing baselines 05.11 and 05.12.

For consistency, these parameters were applied to all images used in the analysis, including products for which XML metadata were unavailable. Consequently, the radiometric parameters were not independently verified for every processed image.

Under this assumption, reflectance was calculated as:

$$\rho_{\mathrm{BOA}} = \frac{DN - 1000}{10000}. \tag{2}$$

The Normalized Difference Vegetation Index (NDVI) was calculated from the corrected red and near-infrared reflectance values:

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{B08}} - \rho_{\mathrm{B04}}}{\rho_{\mathrm{B08}} + \rho_{\mathrm{B04}}}, \tag{3}$$

where ρ_{B08} represents near-infrared BOA reflectance and ρ_{B04} represents red BOA reflectance.

The additive radiometric offsets were applied before NDVI calculation. Although a common multiplicative factor cancels in the normalized difference, an additive offset does not. Consequently, the treatment of the additive offset affects the calculated NDVI values.

Pixels containing zero digital numbers, nonfinite values, negative corrected reflectance, or nonpositive reflectance sums were excluded. Calculated NDVI values outside the theoretical interval $[-1, 1]$ were also rejected.

Invalid pixels were assigned a NoData value of -9999. The resulting NDVI rasters were exported as single-band, 32-bit floating-point GeoTIFF files.

4.6 NDVI Mosaicking and Spatial Extraction

NDVI was initially calculated independently for each complete Sentinel-2 B04/B08 tile pair.

For acquisition dates with both required tiles, the corresponding tile-level NDVI rasters from T17QLF and T17QMF were mosaicked to provide spatial coverage across the analysis region.

The agricultural analysis region was defined using five polygons from the integrated soil–vegetation dataset.

Four polygons were associated with Ferralítico Rojo Típico (Typical Red Ferrallitic Soil), and one was associated with Ferralítico Amarillento Típico (Typical Yellowish Ferrallitic Soil).

The selected polygons were combined into a unified analysis geometry covering approximately 16.22 km^2. This geometry was used consistently for regional NDVI extraction across the selected Sentinel-2 acquisitions.

The geometry of the selected agricultural analysis region was used to define the extraction area. The relevant vector geometry was transformed from EPSG:3795 to EPSG:32617, the coordinate reference system of the Sentinel-2 rasters, before spatial extraction.

The mosaics were cropped to the analysis geometry and exported as GeoTIFF files at 10 m spatial resolution.

The processing inventory contained 64 tile-level NDVI rasters and 32 analysis-area NDVI rasters, corresponding to 32 acquisition dates. These counts describe the processed inventory before the final regional cloud screening used to select the 15 acquisitions for temporal analysis.

4.7 QGIS-Based Selection of Potato Fields

QGIS was used to identify three potato fields within the processed agricultural region. The selected locations were verified using local knowledge of the study area and were used for a more focused examination of vegetation-index variability.

The sampling areas were digitized as three separate circular polygons and overlaid on the processed NDVI rasters for visual inspection. These polygons represent selected sampling areas rather than surveyed field boundaries.

The same three sampling polygons were used to extract NDVI observations for the available acquisition dates, allowing temporal comparisons within and between the sampled areas.

The field-level dataset contained 45 field-date observations, corresponding to three sampling areas evaluated over 15 Sentinel-2 acquisition dates. For each sampling area and date, the extracted data were summarized using the number of valid pixels, mean NDVI, median NDVI, and standard deviation.

The resulting dataset supports comparisons of temporal NDVI patterns among the three sampled areas. Because the sampling polygons represent selected locations rather than a comprehensive survey of potato cultivation, the results should not be interpreted as representative of every potato field in the study region.

4.8 Raster Validation and Temporal Sampling

The analysis-area NDVI rasters were validated to assess their spatial and numerical consistency.

Validation included checks of the coordinate reference system, spatial resolution, raster dimensions, data type, number of bands, NoData representation, and range of valid NDVI values.

The expected raster characteristics were EPSG:32617, 10 m spatial resolution, a single 32-bit floating-point band, and a NoData value of -9999. Valid NDVI values were required to fall within the interval $[-1, 1]$.

A 15-date temporal sampling design was used to examine vegetation-index variability across three broad periods: November–February, March–April, and May–June. Candidate acquisitions were evaluated according to temporal distribution and the availability of suitable observations.

Only dates represented by processed and validated imagery were included in the subsequent temporal analysis.

4.9 Temporal and Agricultural-Region Analysis

The accepted NDVI observations were analyzed to characterize temporal variation in vegetation-index values within the agricultural analysis region and the three selected sampling areas.

Regional NDVI summaries were used to examine changes across acquisition dates. Field-level summaries were used to compare the mean, median, and variability of NDVI within each sampling polygon over the same observation period.

Tables and plots were generated to examine temporal trends and differences among the sampled areas. The analysis was descriptive and focused on the distribution and evolution of satellite-derived NDVI values.

The resulting observations provide information about vegetation-index variability during the study period. They are not direct measurements of crop yield, soil properties, or agricultural management, and they do not independently establish the causes of observed temporal changes.

5 Results

The GeoSoil-Cuba workflow produced validated spatial datasets, satellite-derived vegetation indices, and statistical summaries describing soil-map coverage and vegetation variability within the study area.

The results are organized into eight subsections covering soil topology, spatial coverage, satellite acquisition screening, preliminary NDVI inspection, agricultural-region selection, temporal variability, spatial NDVI distributions, and field-level observations.

5.1 Soil Topology and Area-Accounting Validation

The topology assessment identified differences between the summed areas of individual clipped soil polygons and the area of their geometric union. These measurements were compared in the source coordinate reference system (EPSG:3795) and after reprojection to UTM Zone 17N (EPSG:32617).

In EPSG:3795, the sum of all clipped soil polygon areas was 162.907 km^2, comprising 161.984 km^2 of classified soil polygons and 0.922 km^2 associated with an undefined classification. The corresponding geometric union covered 162.852 km^2, yielding a difference of approximately 0.055 km^2.

After reprojection to EPSG:32617, the sum of all polygon areas decreased to 162.820 km^2, comprising 161.899 km^2 of classified soil polygons and 0.922 km^2 associated with the undefined classification. The geometric union covered 162.766 km^2, again yielding a difference of approximately 0.055 km^2.

The total polygon-area sum decreased by 0.086 km^2 (approximately 0.053%) between the two CRSs. The geometric union area decreased by a similar amount, while the difference between the polygon-area sum and the geometric union remained nearly unchanged. These results demonstrate the sensitivity of planar area measurements to coordinate reference system transformations, while confirming that the overlap-related area difference was consistent between the two CRSs.

Pairwise intersection analysis identified 90 polygon pairs with positive-area overlaps. Their cumulative pairwise overlap area was 0.054919 km^2, representing approximately 0.034% of the summed polygon area in EPSG:32617.

The largest individual overlap measured 0.046807 km^2 and occurred between Aluvial Poco Diferenciado (2618) and Pardo con Carbonatos Típico (1001).

The principal area-accounting and topology results are summarized in Table 2.

Table 2: Soil polygon area accounting and topology validation results.

Metric	Value
Area measurements in EPSG:3795	
Classified soil polygon-area sum (km^2)	161.984
Undefined soil polygon area (km^2)	0.922
Total polygon-area sum (km^2)	162.907
Geometric union area (km^2)	162.852
Sum minus union (km^2)	0.055
Area measurements in EPSG:32617	
Classified soil polygon-area sum (km^2)	161.899
Undefined soil polygon area (km^2)	0.922
Total polygon-area sum (km^2)	162.820
Geometric union area (km^2)	162.766
Sum minus union (km^2)	0.055
CRS comparison	
Change in polygon-area sum (km^2)	−0.086
Relative change in polygon-area sum (%)	−0.053
Topology validation	
Overlapping polygon pairs	90
Total pairwise overlap area (km^2)	0.054919
Largest individual overlap (km^2)	0.046807
Pairwise overlap / total polygon area (%)	0.034

The cumulative pairwise overlap area closely matched the difference between the summed polygon areas and their geometric union in EPSG:32617. Specifically, the sum-minus-union difference was 0.054919 km^2, matching the reported cumulative pairwise overlap area to six decimal places.

However, pairwise overlap calculations may count the same location more than once where three or more polygons intersect. Consequently, the cumulative pairwise overlap area should not be interpreted as an exact measure of the geographic area affected by overlaps. Likewise, the difference between the polygon-area sum and the geometric union represents excess area arising from overlapping polygons rather than necessarily the unique geographic extent of the overlapping regions.

The detected overlaps were retained as data-quality findings. No source soil-unit boundaries were modified or removed to eliminate these intersections.

5.2 Soil and Vegetation Spatial Coverage

The Area of Interest (AOI) covers approximately 178.47 km^2. The spatial coverage assessment reported soil-map coverage of approximately 162.85 km^2, equivalent to 91.25% of the AOI.

The corresponding uncovered area was 15.62 km^2, representing approximately 8.75% of the AOI.

Water bodies occupied approximately 14.49 km^2 within the AOI. Of this area, approximately 3.07 km^2 overlapped existing soil polygons, while 11.42 km^2 was not covered by the soil dataset.

The remaining non-water gaps accounted for approximately 4.20 km^2, equivalent to 2.35% of the AOI.

The principal spatial coverage measurements are presented in Table 3.

Table 3: Reported spatial coverage of soil, water-body, and vegetation datasets within the Area of Interest.

Spatial component	Area (km^2)	AOI (%)
Area of Interest	178.47	100.00
Soil-map coverage	162.85	91.25
Total soil-data gap	15.62	8.75
Water-body area	14.49	8.12
Water area not covered by soil polygons	11.42	6.40
Remaining non-water gaps	4.20	2.35
Vegetation coverage	161.93	90.73

The soil-map coverage reported in Table 3 differs slightly from the geometric union area reported in Table 2.

The two measurements are therefore retained as distinct reported results. Their difference requires verification against the processing stages and geometry sets used to calculate the respective areas.

The spatial distribution of the remaining soil-data gaps is shown in Figure 3.

The clipped soil inventory contained one polygon with the attribute combination `Cód_tipo = 0` and `Cód_stipo = 0`.

Because code zero was not defined as a valid soil type or subtype in Anexo 6, the polygon was classified as undefined rather than assigned to an unsupported soil category.

This polygon accounted for 0.922 km^2 of the summed soil polygon area.

The spatial distribution of the mapped soil units is presented in Figure 4.

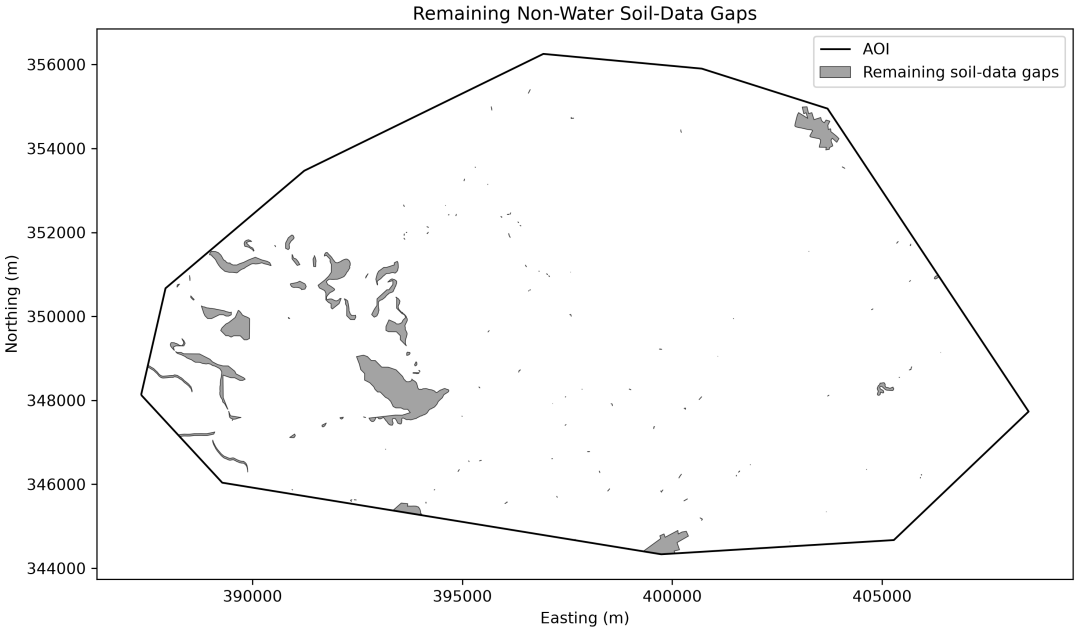


Figure 3: Spatial distribution of remaining areas without soil-map coverage within the AOI, following the identification of water-related gaps associated with the Mampostón reservoir.

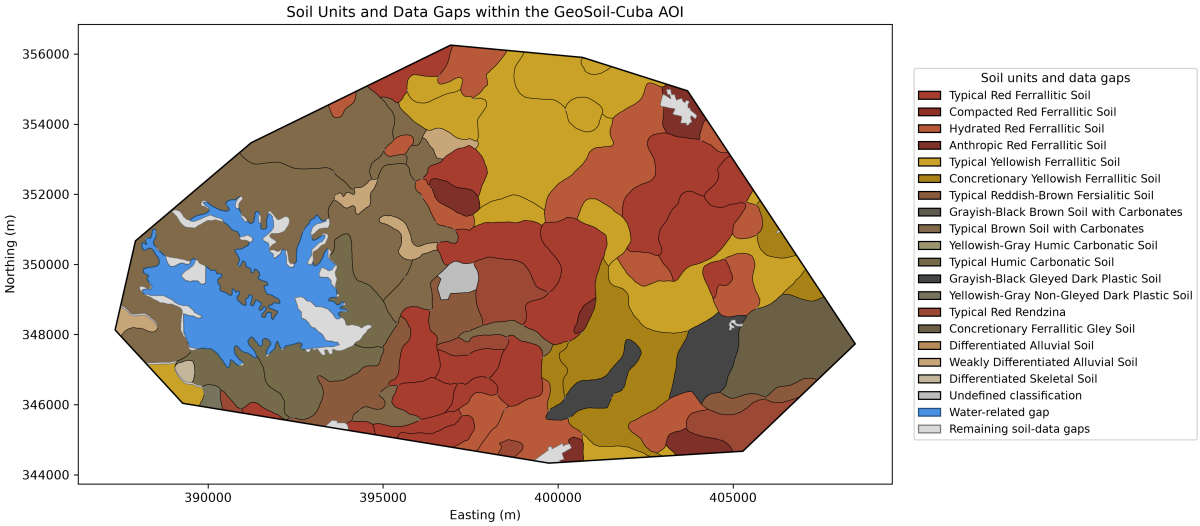


Figure 4: Spatial distribution of mapped soil units within the AOI, including areas with undefined soil classification.

The vegetation and land-cover dataset provided reported spatial coverage of approximately 161.93 km^2.

The inventory included grasslands (87.45 km^2), cultivated land (75.86 km^2), secondary vegetation (13.69 km^2), and flooded hydromorphic areas (1.47 km^2).

The sum of these reported class areas exceeds the stated vegetation coverage. Consequently, these measurements cannot be interpreted as a mutually exclusive partition of the vegetation-covered area without further verification of the classification and area-accounting methods.

The spatial distribution of the vegetation categories is illustrated in Figure 5.

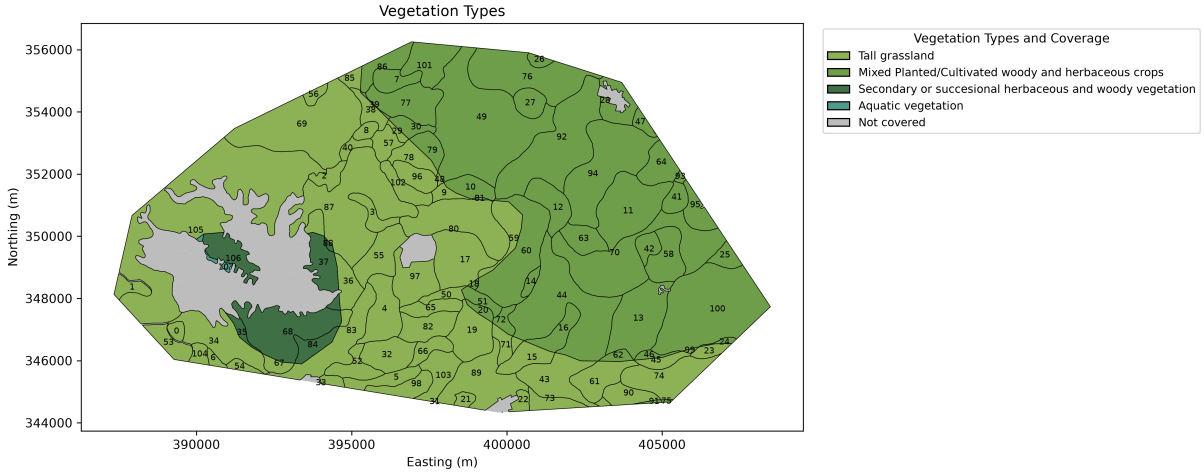


Figure 5: Spatial distribution of mapped vegetation and land-cover categories within the AOI.

These spatial datasets provided the basis for integrating soil classifications with existing vegetation and land-cover information.

5.3 Sentinel-2 Acquisition and Quality Screening

The Sentinel-2 inventory contained 226 observations acquired between November 2025 and June 2026.

The mean catalogue-reported cloud-cover percentage was 40.04%, indicating substantial variation in acquisition conditions across the available imagery.

Following satellite processing, 32 NDVI rasters were generated for the selected agricultural region.

The regional quality-screening procedure retained 15 acquisitions with less than 10% cloud coverage within the analysis region.

The accepted observations spanned 3 November 2025 to 16 June 2026.

The temporal distribution of these acquisitions was not uniform, reflecting the availability of observations meeting the regional screening criterion.

Fourteen retained rasters contained 162,195 valid NDVI pixels, while one contained 162,193 valid pixels.

The nearly constant valid-pixel counts indicate that the accepted rasters represented a stable spatial sampling extent, although this numerical consistency does not independently establish the absence of residual cloud or shadow contamination.

5.4 Initial Inspection of NDVI Data

A preliminary inspection was conducted using 14 NDVI rasters from the available Sentinel-2 observations.

The initial analysis examined spatial NDVI patterns, temporal changes in summary statistics, and differences in pixel-level distributions.

A representative NDVI map from 3 November 2025 is presented in Figure 6.

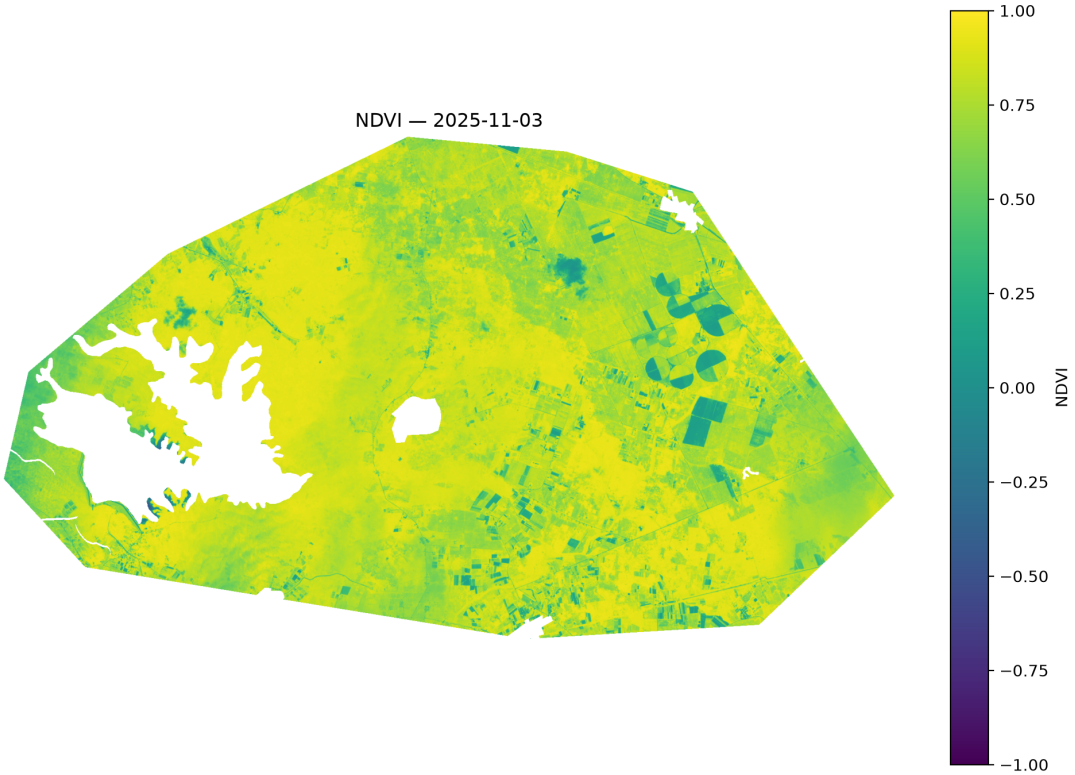


Figure 6: Preliminary NDVI map derived from Sentinel-2 imagery acquired on 3 November 2025.

The initial temporal summaries are shown in Figure 7, while the corresponding pixel-level distributions are presented in Figure 8.

Inspection revealed that some rasters provided incomplete spatial coverage of the broader study area.

The incomplete coverage was associated with limitations in the available Sentinel-2 imagery rather than with the mapped reservoir.

The subsequent agricultural analysis region was located outside the affected areas, allowing a consistent spatial extent to be used for the final temporal analysis.

The preliminary observations are therefore presented as exploratory results and are distinguished from the final 15-acquisition analysis.

5.5 Selection of the NDVI Analysis Region

Five agricultural polygons were selected from the integrated soil–vegetation dataset and combined into a unified analysis region.

Four polygons were associated with Ferralítico Rojo Típico (Typical Red Ferrallitic Soil), while one was associated with Ferralítico Amarillento Típico (Typical Yellowish Ferrallitic Soil).

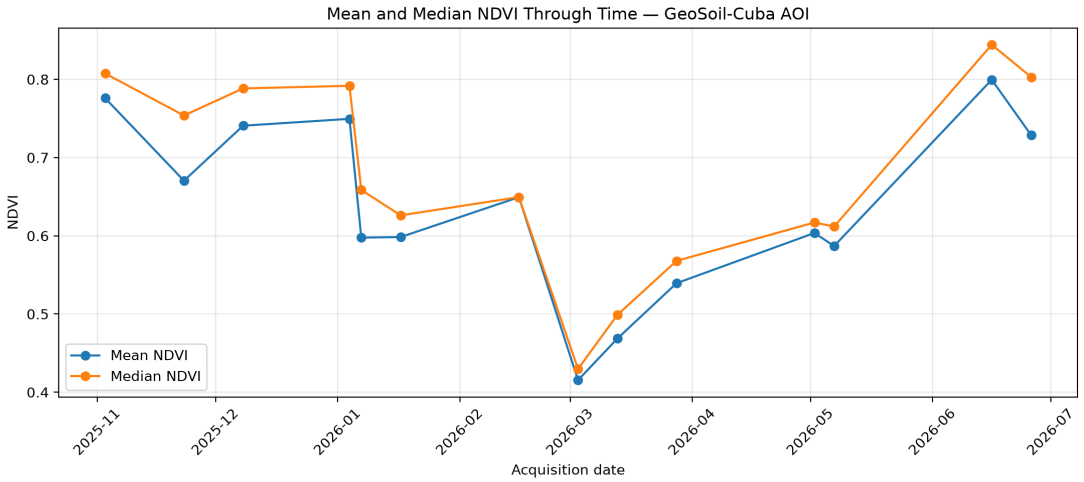


Figure 7: Preliminary temporal variation in mean and median NDVI across the initially selected raster observations.

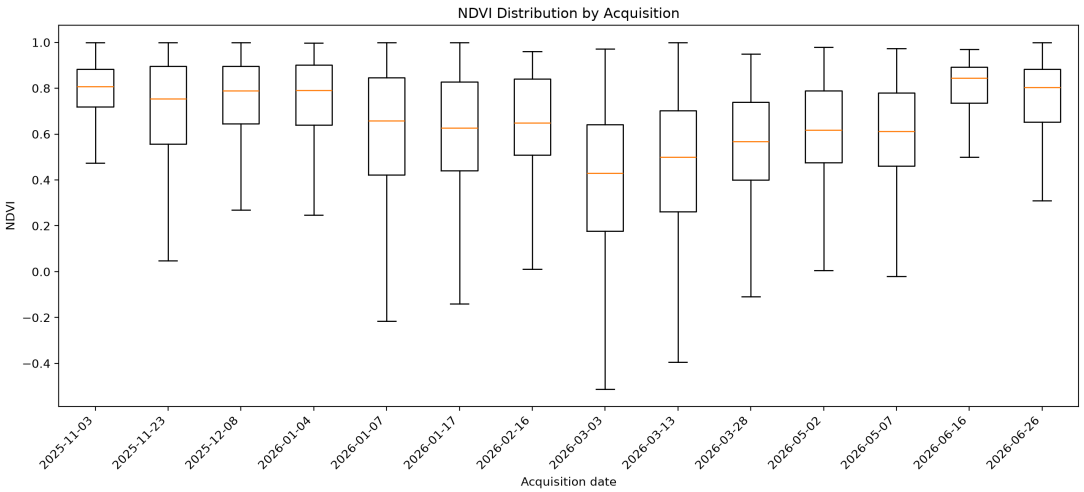


Figure 8: Preliminary distributions of pixel-level NDVI values across the initially selected Sentinel-2 observations.

The resulting agricultural analysis region covered approximately 16.22 km^2.

Its spatial distribution is presented in Figure 9.

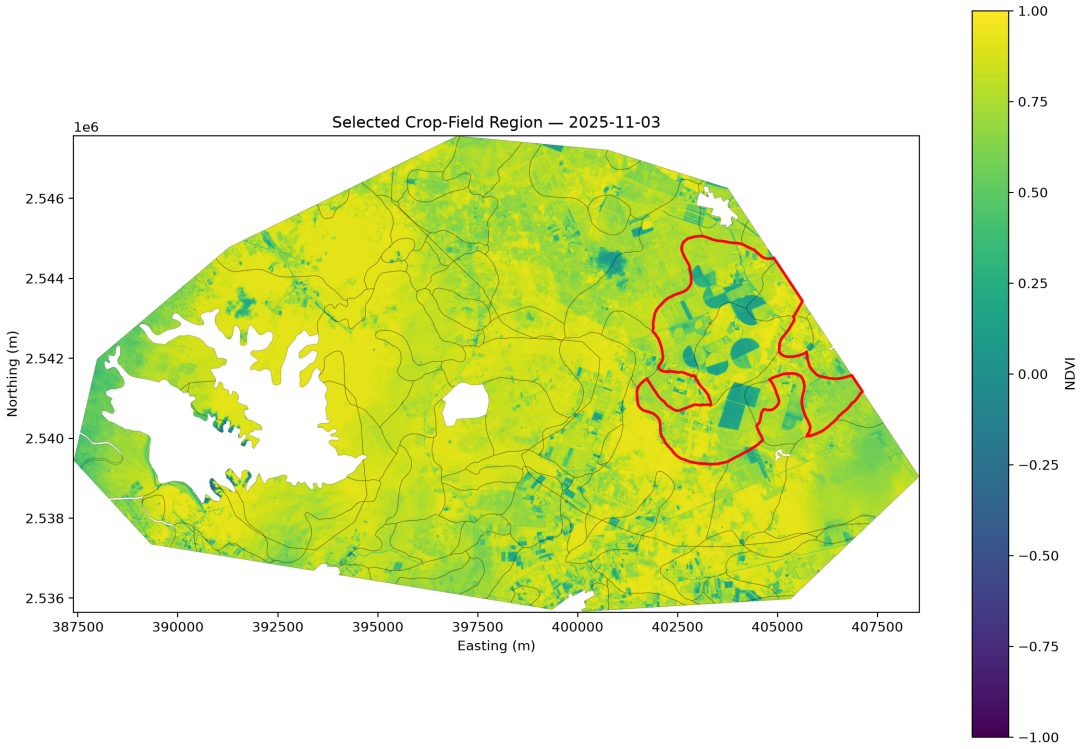


Figure 9: Spatial distribution of the five selected agricultural polygons and their unified analysis region within the study area.

The selected region was located outside the areas affected by incomplete satellite coverage identified during the preliminary inspection.

The 32 regional NDVI rasters provided the initial image collection for this area.

Following regional SCL-based quality screening, 15 acquisitions were retained for the temporal and spatial NDVI analyses.

The retained acquisitions are illustrated in Figure 10.

The unified region provided a consistent geographic basis for comparing vegetation-index observations throughout the study period.

5.6 Temporal NDVI Variability

Temporal NDVI variability was examined using the mean and median values calculated for the 15 accepted Sentinel-2 acquisitions.

The results indicate a general decline in regional NDVI from November 2025 through March 2026.

Mean NDVI decreased from approximately 0.688 on 3 November 2025 to 0.466 on 15 March 2026.

Over the same period, median NDVI declined from approximately 0.754 to 0.463.

Subsequent observations showed fluctuations rather than a continuous increase.

Mean NDVI was approximately 0.470 on 1 April 2026, 0.512 on 29 April, and 0.482 on 7 May.

By 16 June 2026, mean NDVI had increased to approximately 0.737, while median NDVI reached approximately 0.764.

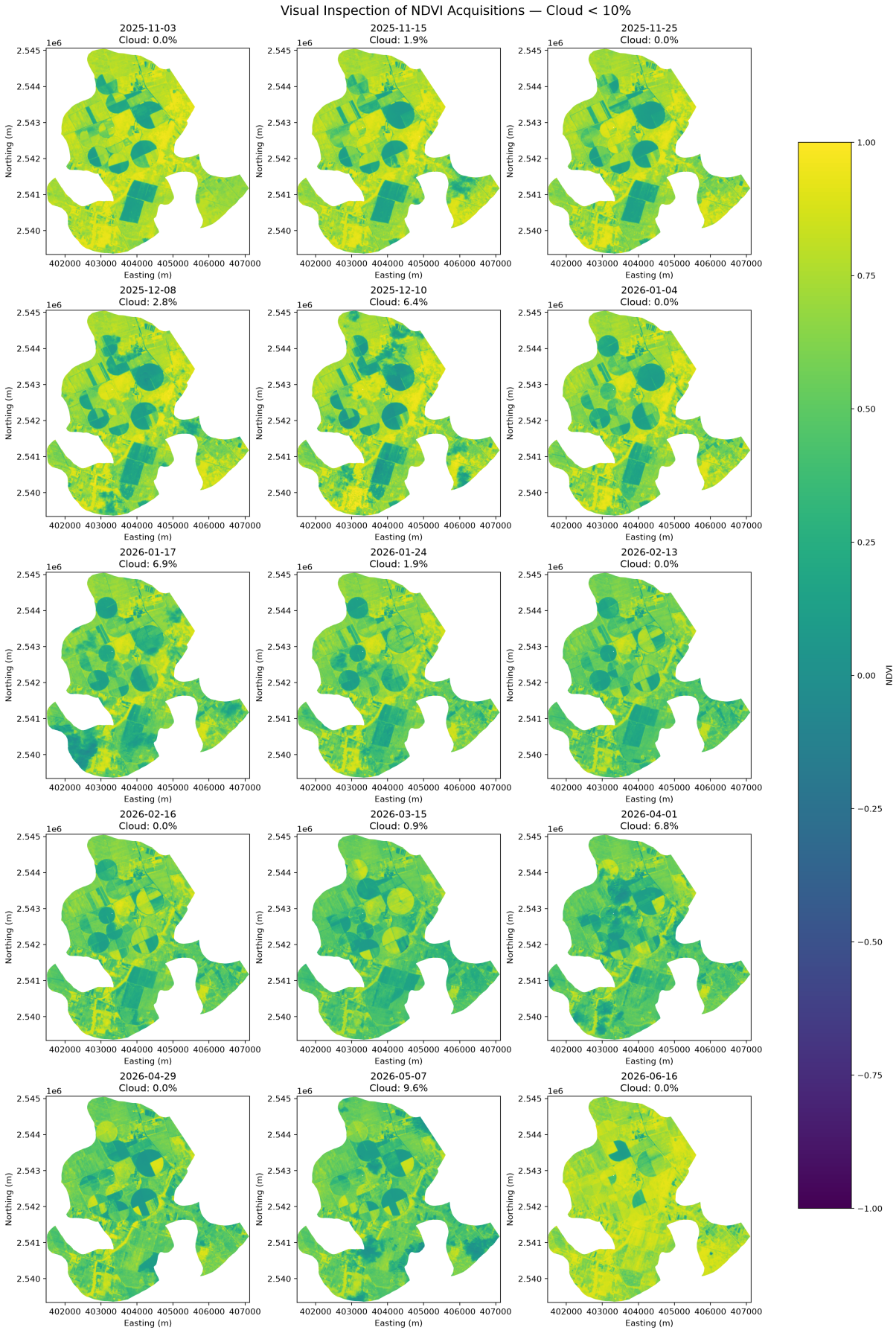


Figure 10: Visual comparison of the 15 regional NDVI acquisitions satisfying the cloud-coverage criterion of less than 10%.

The temporal evolution of both statistics is presented in Figure 11.

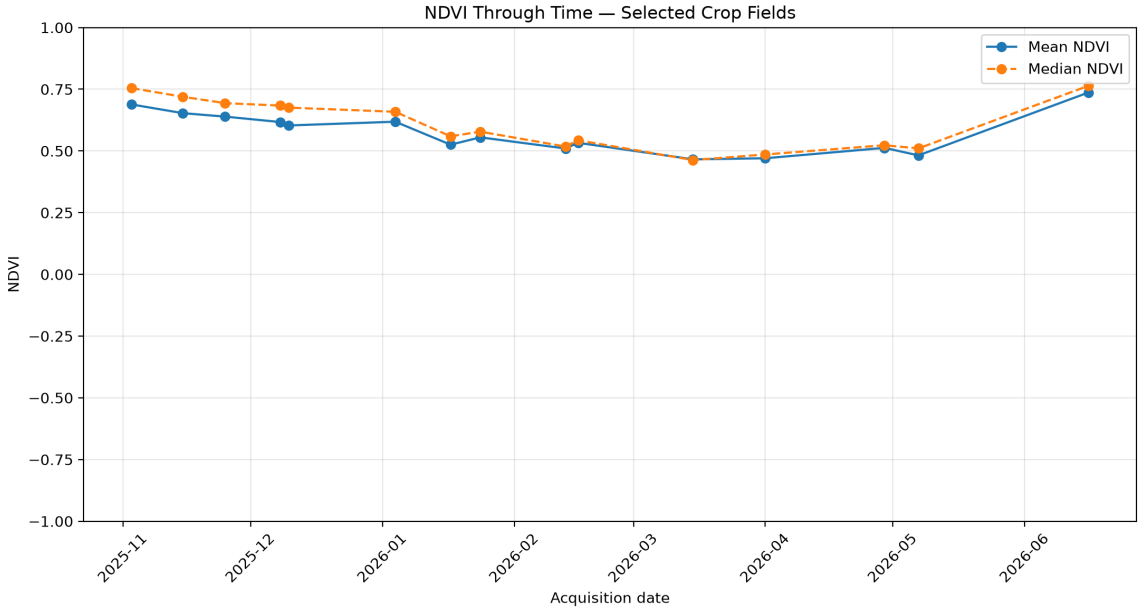


Figure 11: Temporal evolution of mean and median NDVI across the selected agricultural region for the 15 accepted Sentinel-2 acquisitions from November 2025 to June 2026.

The mean and median series exhibited broadly similar temporal patterns.

However, their differences at individual acquisition dates indicate that the distributions were not necessarily symmetric throughout the observation period.

The regional statistics describe changes in satellite-derived vegetation-index values and do not independently establish the agricultural or environmental processes responsible for them.

5.7 NDVI Distribution and Spatial Variability

Pixel-level NDVI distributions were examined to characterize variability within the selected agricultural region.

Boxplots for the 15 accepted acquisitions are presented in Figure 12.

The distributions showed changes in median NDVI, interquartile range, and overall dispersion across acquisition dates.

A general shift toward lower NDVI values was observed between November 2025 and March 2026, consistent with the regional mean and median time series.

To examine differences in distribution shape, four representative acquisitions were selected:

8 December 2025, 16 February 2026, 29 April 2026, and 16 June 2026.

Their NDVI density distributions are shown in Figure 13.

The density curves showed differences in the relative frequency of low, intermediate, and high NDVI values.

These changes indicate that temporal variability was not limited to shifts in the regional mean.

The spatial distribution of NDVI was also examined for the same four acquisitions, as shown in Figure 14.

The maps revealed geographic differences in NDVI within the agricultural region.

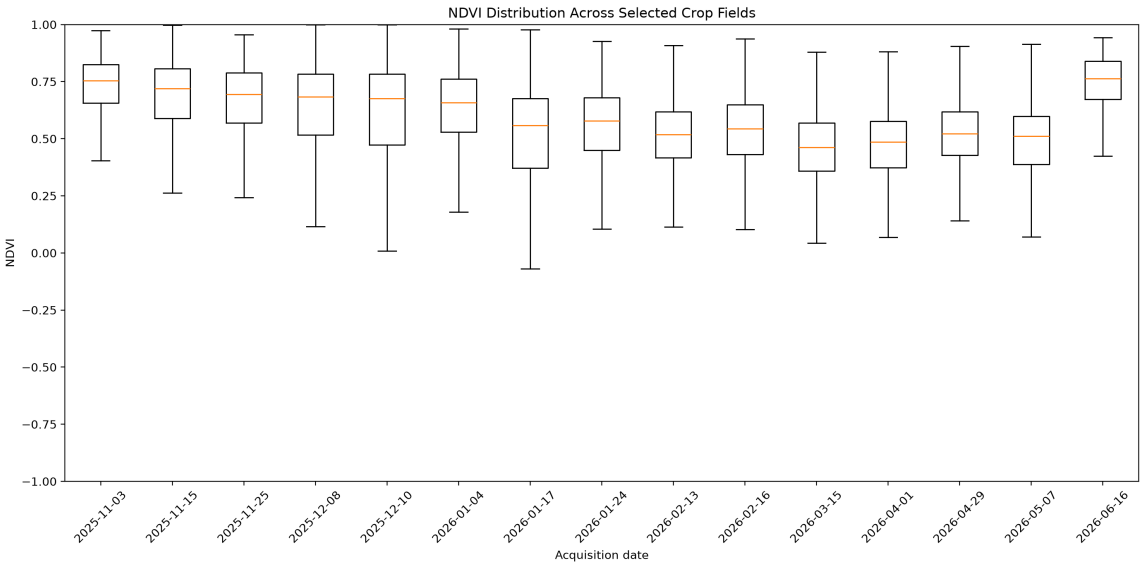


Figure 12: Pixel-level NDVI distributions across the 15 accepted Sentinel-2 acquisitions, represented by boxplots.

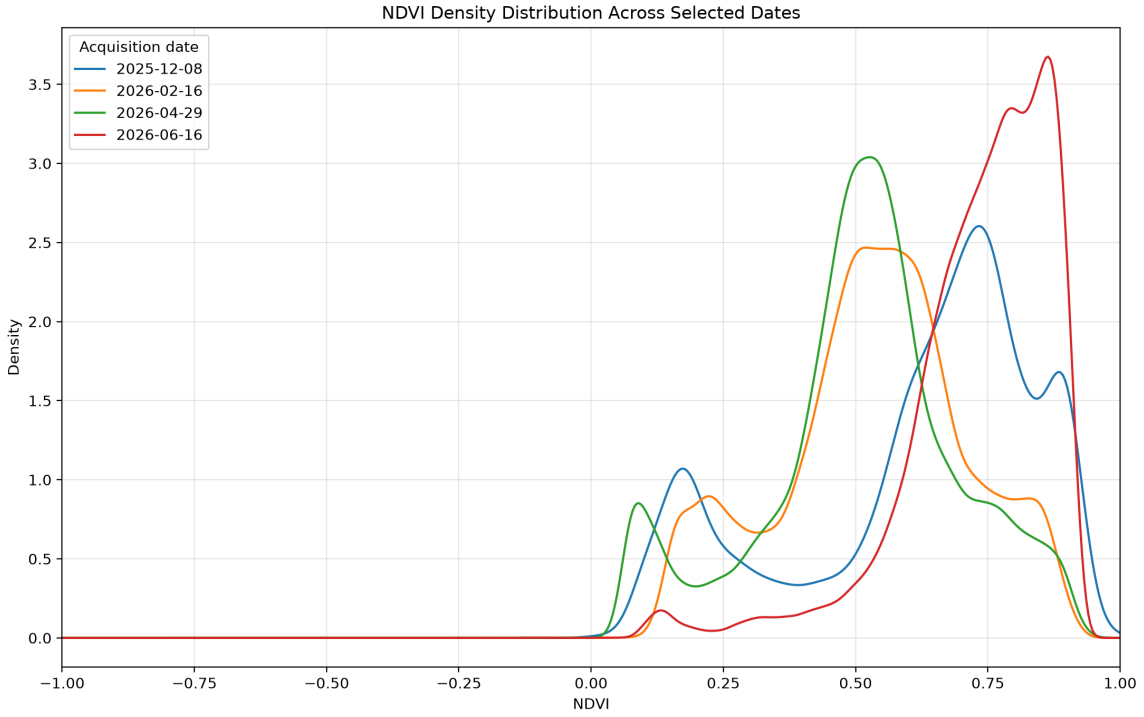


Figure 13: Pixel-level NDVI density distributions for four representative acquisitions: 8 December 2025, 16 February 2026, 29 April 2026, and 16 June 2026.

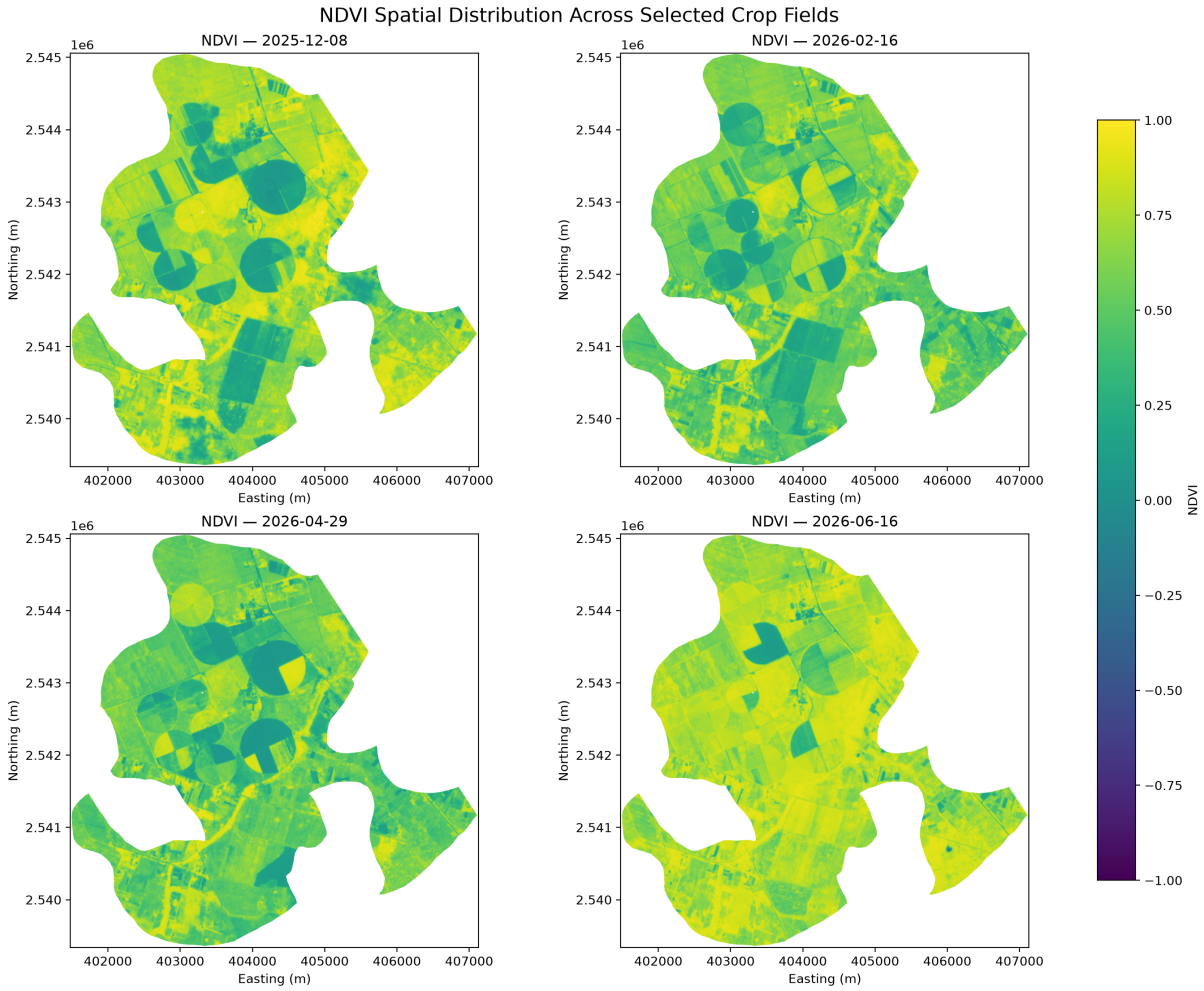


Figure 14: Spatial distribution of NDVI within the selected agricultural region on 8 December 2025, 16 February 2026, 29 April 2026, and 16 June 2026. All panels use the same NDVI scale.

Relatively high and low vegetation-index values were not distributed uniformly across the selected polygons.

The June observation exhibited higher regional NDVI values than those recorded during March and April.

However, these regional distributions include multiple agricultural and vegetation conditions and cannot be interpreted as observations of potato cultivation alone.

5.8 Field-Level NDVI Analysis of Selected Potato Fields

A complementary field-level analysis examined three locally verified potato fields within the agricultural region.

The fields were represented by circular sampling polygons digitized in QGIS.

Their locations and the sampling configuration are illustrated in Figure 15.

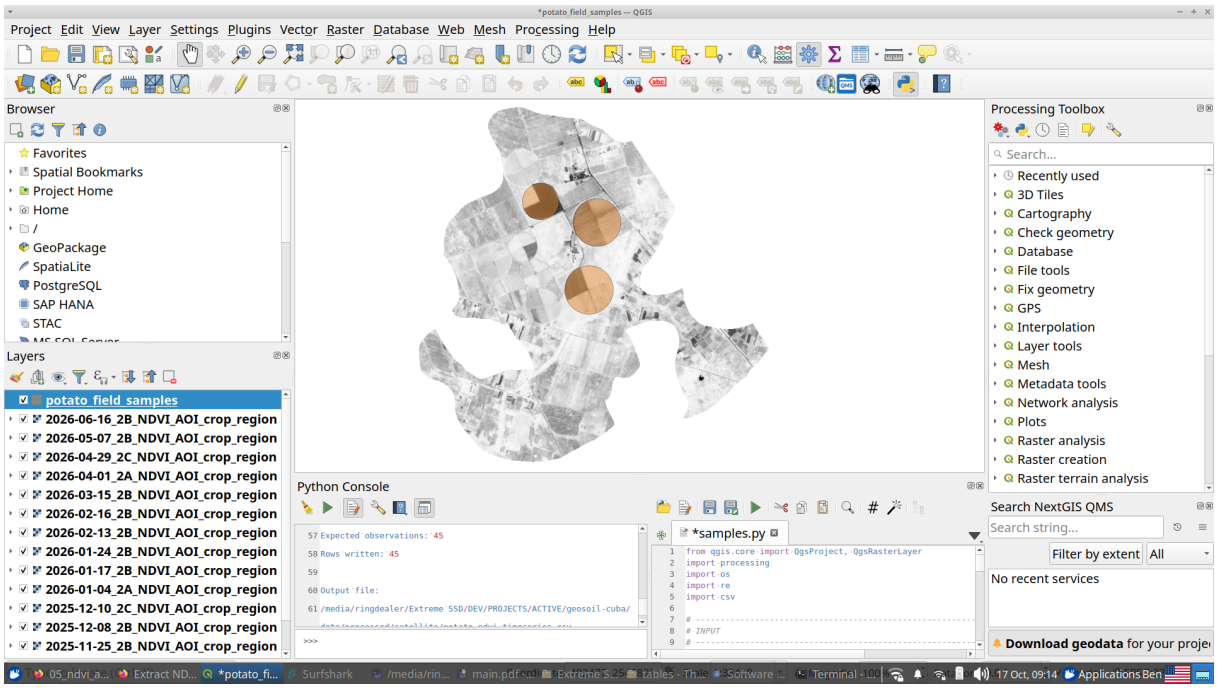


Figure 15: Three locally verified potato-field sampling areas digitized in QGIS and displayed over the agricultural-region NDVI raster.

NDVI observations were extracted for each sampling polygon across the 15 accepted acquisition dates, yielding 45 field-date observations.

The number of valid pixels differed among the sampling polygons but remained constant across the acquisition dates for each field.

Field 1 contained 4,603 valid pixels, Field 2 contained 4,785 pixels, and Field 3 contained 2,710 pixels.

The temporal evolution of mean NDVI for the three fields is presented in Figure 16.

The three sampled areas exhibited different temporal NDVI patterns.

Field 1 reached its highest mean NDVI of approximately 0.744 on 15 March 2026.

Field 2 reached its highest mean NDVI of approximately 0.690 on 16 June 2026.

Field 3 reached its highest mean NDVI of approximately 0.573 on 24 January 2026, followed by a substantial decline in subsequent observations.

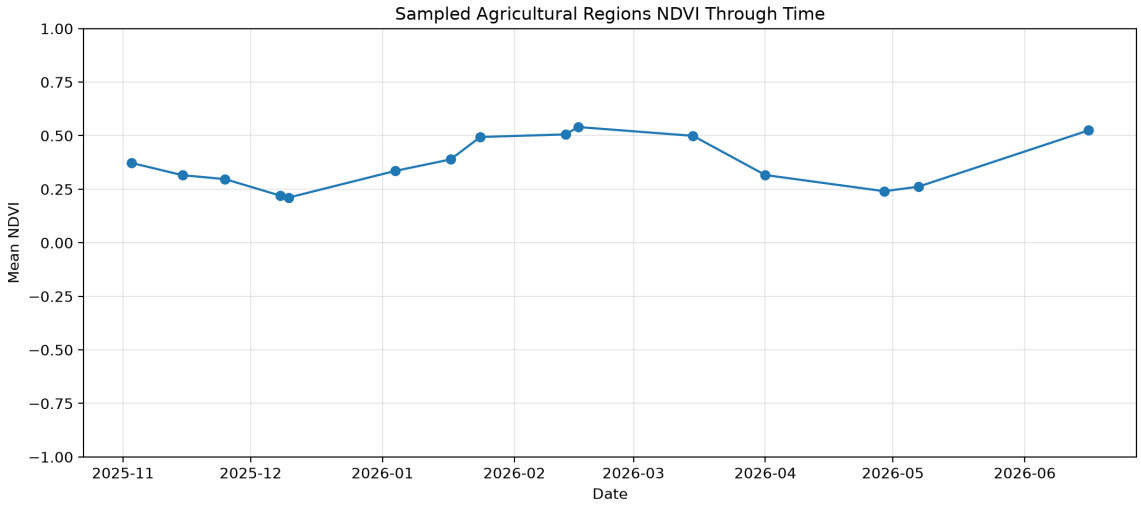


Figure 16: Temporal evolution of mean NDVI across the three selected potato-field sampling areas from November 2025 to June 2026.

The maximum observed mean NDVI values and their corresponding dates are summarized in Table 4.

Table 4: Maximum observed mean NDVI for the three selected potato-field sampling areas.

Field	Maximum mean NDVI	Acquisition date
Field 1	0.744	15 March 2026
Field 2	0.690	16 June 2026
Field 3	0.573	24 January 2026

The different dates of maximum mean NDVI indicate that the sampled fields did not follow identical vegetation-index trajectories.

Pixel-level NDVI distributions were also examined using boxplots, as presented in Figure 17.

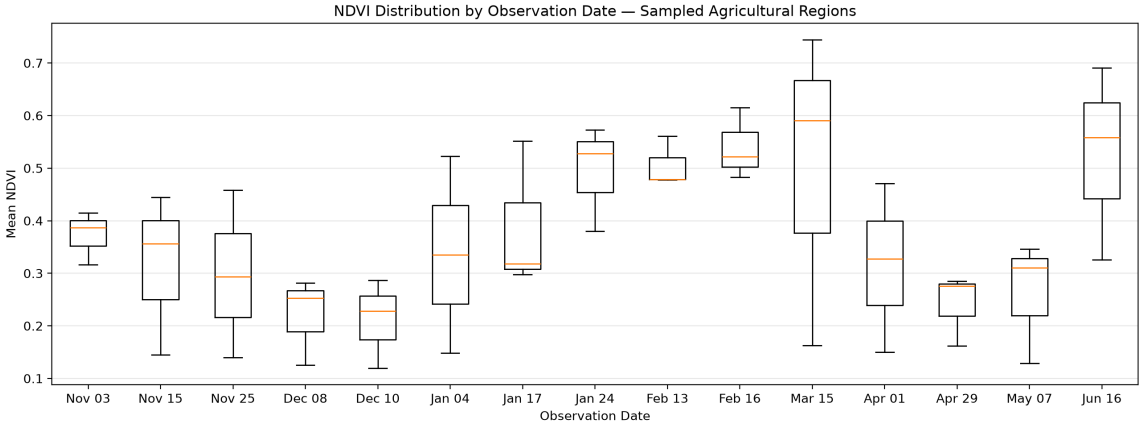


Figure 17: Distribution of NDVI observations across the three selected potato-field sampling areas and acquisition dates.

The field-level results demonstrate temporal variability within the selected agricultural sampling areas that is not fully represented by the regional mean NDVI series.

The observed differences provide a basis for further investigation of agricultural conditions but do not independently identify crop-development stages, harvesting dates, or management

practices.

6 Discussion and Limitations

6.1 Spatial Data Integration and Map Completeness

The results demonstrate the importance of evaluating geometric consistency and spatial coverage before using historical soil cartography in contemporary GIS analysis. Within the 178.47 km^2 Area of Interest (AOI), the soil polygons covered approximately 162.852 km^2, measured as their geometric union in EPSG:3795. Of the 15.62 km^2 not covered by soil polygons, approximately 11.42 km^2 corresponded to mapped water surfaces, leaving 4.20 km^2 of non-water gaps.

Distinguishing water surfaces from unresolved gaps prevents the absence of terrestrial soil polygons over the reservoir from being incorrectly interpreted as a soil-mapping defect. The remaining gaps require examination against the original cartography and other authoritative spatial sources before they can be classified or filled.

Topology checks identified 90 polygon pairs with positive-area intersections. Their cumulative pairwise overlap area was approximately 0.054919 km^2. Although small relative to the mapped extent, these overlaps can introduce duplicate area contributions in spatial summaries and overlay operations.

The original geometries were preserved to maintain data provenance. Consequently, the processed dataset should not be described as fully topologically corrected. Area statistics based on sums of individual polygons must be distinguished from statistics based on their geometric union. Furthermore, cumulative pairwise overlap areas do not necessarily equal the geographic area affected by overlaps when three or more polygons intersect.

Area calculations also varied slightly with the coordinate reference system (CRS). In EPSG:3795, the sum of individual polygon areas was 162.907 km^2, while their geometric union covered 162.852 km^2. After reprojection to EPSG:32617, the corresponding values were 162.820 km^2 and 162.766 km^2, respectively. The union area decreased by approximately 0.086 km^2 (0.053%) after reprojection. In both CRSs, the difference between the polygon-area sum and the geometric union remained approximately 0.055 km^2, consistent with the contribution of overlapping polygons to the summed area. These results emphasize the need to specify the CRS and area-calculation method when reporting spatial coverage.

The integration of soil and vegetation polygons provides a common spatial framework for examining mapped attributes. Nevertheless, spatial coincidence does not establish that a soil class determines the observed vegetation condition. The source layers may differ in production date, mapping purpose, classification criteria, and positional accuracy. These differences limit the interpretation of direct spatial comparisons.

An additional area-accounting issue concerns the vegetation dataset. The four reported vegetation class areas sum to approximately 178.47 km^2, whereas the stated vegetation coverage is 161.93 km^2. Their mutual exclusivity, possible overlaps, and the corresponding area-calculation procedures must therefore be verified before the categories can be interpreted as an additive partition of the AOI.

6.2 Interpretation of Regional NDVI Dynamics

The 15 retained Sentinel-2 observations show substantial temporal variation within the selected 16.22 km^2 agricultural region.

Mean NDVI declined from approximately 0.688 on 3 November 2025 to 0.466 on 15 March 2026, then varied during April and May before reaching approximately 0.737 on 16 June 2026.

The boxplots, density curves, and spatial maps indicate that these changes involved both regional central tendency and the spatial distribution of NDVI values.

These observations are consistent with changes in vegetation conditions across an actively managed agricultural landscape, but they do not independently identify their causes.

The selected region contains multiple agricultural polygons rather than an independently verified single-crop planting area. Consequently, its aggregate trajectory cannot be interpreted as a potato growth curve.

Changes in crop composition, planting and harvesting schedules, vegetation regrowth, weeds, moisture conditions, and residual image contamination could contribute to the observed signal.

In particular, the June increase cannot be attributed to a specific crop or management practice without additional field evidence.

The regional analysis is descriptive rather than causal. The association of the selected polygons with mapped Ferrallitic soil units provides geographic context, but the analysis does not control for irrigation, agricultural management, crop identity, or other potential confounding factors.

The results therefore do not support claims that one mapped soil class produced higher NDVI or better crop performance than another.

6.3 Field-Level Observations and Agricultural Context

The three locally verified potato fields provide a more specific spatial reference than the broader agricultural region.

Their NDVI trajectories were not uniform. Among the retained observations, the highest mean NDVI occurred on 15 March 2026 in Field 1, on 16 June 2026 in Field 2, and on 24 January 2026 in Field 3.

These differences illustrate why an aggregate regional NDVI curve should not be interpreted as representative of every individual field.

However, the dates of the highest observed mean NDVI values do not necessarily represent the actual seasonal vegetation maxima. Unobserved peaks may have occurred between the available satellite acquisitions.

Local verification supports the identification of the sampling locations as potato fields, but it does not establish that potatoes occupied each sampled polygon continuously throughout the November 2025–June 2026 observation period.

In particular, post-harvest NDVI may reflect subsequent crops, vegetation regrowth, weeds, or other vegetation rather than the original potato crop.

The circular sampling polygons are approximations rather than surveyed field boundaries. Their geometry and placement may therefore influence the extracted NDVI statistics, particularly where sampling areas intersect heterogeneous vegetation or field boundaries.

Field-specific planting dates, harvesting records, accurate field boundaries, and repeated ground observations would be needed to assign crop-development stages and interpret field-level

differences with greater confidence.

6.4 Remote-Sensing and Reproducibility Limitations

The regional quality filter retained acquisitions with less than 10% cloud coverage according to the selected SCL cloud classes.

This acquisition-level threshold reduces potential contamination but does not guarantee that every pixel used for NDVI statistics is free from clouds, cirrus, haze, or shadows.

Cloud shadows were evaluated separately, and numerical NDVI validity alone does not constitute a cloud mask.

Future analyses should implement and document explicit pixel-level quality masks, including cloud shadows and other unsuitable SCL classes, and report the number and proportion of usable pixels for each acquisition.

The temporal sample is also irregular. Fifteen retained dates span November 2025 through June 2026, with unequal intervals between observations.

These data support comparisons among the retained acquisition dates but do not provide a continuously observed vegetation-development curve.

Short-lived changes and seasonal maxima may have occurred between acquisitions. Furthermore, the nearly constant number of numerically valid pixels does not independently establish equivalent atmospheric quality across the observations.

Radiometric preprocessing used a quantification factor of 10,000 and an additive offset of -1000 for both spectral bands, based on inspection of 111 available product metadata files.

These parameters were also applied to some images without individually available XML metadata.

Although this assumption was documented, product-specific verification remains necessary to establish the consistency of radiometric preprocessing across the full time series.

Unlike a common multiplicative factor, an additive radiometric offset does not cancel during NDVI calculation. Consequently, incorrect offset parameters may affect the resulting vegetation-index values.

Additional sources of uncertainty include the 10 m spatial resolution of the spectral bands, the 20 m resolution of the SCL, resampling choices, and potential positional differences between historical vector boundaries and satellite imagery.

Overall, the principal contribution of GeoSoil-Cuba is a documented and reproducible workflow for integrating historical soil cartography, existing land-cover information, and satellite-derived vegetation indices.

The results support spatial data-quality assessment and exploratory vegetation monitoring, but they do not independently validate soil properties, crop yield, or causal relationships between soil classes and agricultural performance.

The principal priorities for further development are targeted topology review, harmonized area accounting, product-specific radiometric verification, explicit pixel-level quality masking, and field observations aligned with satellite acquisition dates.

7 Conclusions and Future Work

GeoSoil-Cuba demonstrates an open-source geospatial workflow for preparing historical soil cartography, integrating soil and vegetation attributes, and examining Sentinel-2 NDVI observations within an agricultural landscape in Jaruco, Mayabeque, Cuba. The analysis combines vector-data validation, coordinate reference system harmonization, spatial overlay, satellite-image processing, and temporal vegetation-index analysis within a documented GIS and Python environment.

Within the 178.47 km^2 study area, the soil coverage assessment reported approximately 162.85 km^2 of mapped soil units. Accounting for mapped water surfaces reduced the remaining non-water soil-data gaps to approximately 4.20 km^2, equivalent to 2.35% of the AOI. Topology assessment identified 90 overlapping polygon pairs with a cumulative pairwise overlap area of approximately 0.054919 km^2. These findings highlight the importance of evaluating spatial coverage, geometric consistency, and area-accounting procedures before interpreting historical soil-map information.

Satellite processing produced 32 regional NDVI rasters, of which 15 acquisitions satisfied the regional cloud-coverage criterion. Within the selected 16.22 km^2 agricultural region, mean NDVI declined between November 2025 and March 2026, followed by fluctuations and an increase by June 2026. A complementary analysis of three locally verified potato fields revealed different temporal NDVI trajectories, illustrating the importance of distinguishing field-level observations from regional aggregates.

The findings characterize spatial and temporal patterns in the available datasets but do not establish crop phenological stages, agricultural productivity, or causal relationships between soil classes and vegetation conditions. Interpretation remains constrained by uncertainties in historical mapping, unresolved topology and area-accounting discrepancies, irregular satellite sampling, residual cloud contamination, and incomplete field and radiometric metadata verification.

Future development should prioritize reconciling soil and vegetation area calculations, reviewing overlapping soil polygons and remaining non-water gaps, verifying radiometric parameters for individual satellite products, and implementing explicit pixel-level cloud and shadow masking. Additional improvements should include accurate agricultural field boundaries, planting and harvesting records, and ground observations corresponding to satellite acquisition dates. These developments would provide a stronger basis for evaluating vegetation variability across mapped soil units and agricultural management conditions.

The resulting framework provides a foundation for further geospatial research and potential application in other regions where historical soil cartography, ancillary land-cover data, and contemporary satellite observations must be integrated while preserving data provenance and documenting uncertainty.

Acknowledgments

The author acknowledges the Faculty of Geography, University of Havana, Cuba (Facultad de Geografía, Universidad de La Habana), for providing the soil, water-body, and vegetation-cover datasets used in this study and authorizing their use for research, analysis, and public redistribution.

AI Assistance and Author Responsibility

Generative AI tools were used to assist with the organization, drafting, and revision of this article. The author is responsible for the research workflow, the interpretation of the results, and the final content of the manuscript. The reported methods, quantitative results, and references were reviewed against the available data and source materials.

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